## ESS SENTINEL 2.0

# Component report

2026-09-29T09:25:30.455810+00:00

Source: Synthetic reference dataset

Batch: batch-732c43b47f014f6e

Model: ESS-2026-09-29T050332-ebf09b1e

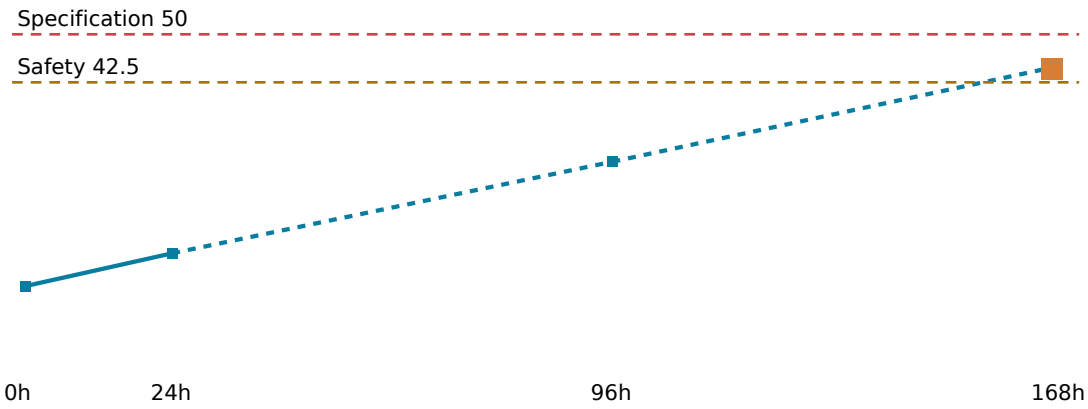
Screening stage: 24h

Decision-support prototype. Synthetic results are not real-world qualification evidence. Final disposition requires qualified engineering review.

## Component disposition

## Leakage current (uA) - CRITICAL

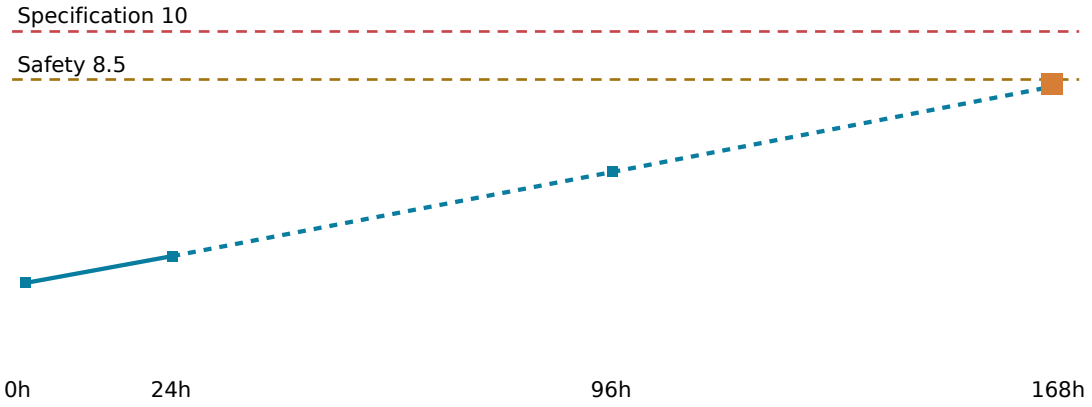
Population anomaly: robust Z=13.48; peer median=10.5 uA, MAD=0.264. Abnormal drift: peer-relative drift Z=18.55; observed change=5.12 uA. Predicted safety crossing: 168h estimate=44.6 uA; upper safety threshold=42.5; margin=-2.1. Isolation Forest score is at training-reference percentile 100.0; this is not a failure probability.




Solid: observed to screening stage. Dashed: retrospective future actual. Orange square: 168h prediction.

## Standby current (mA) - HIGH RISK

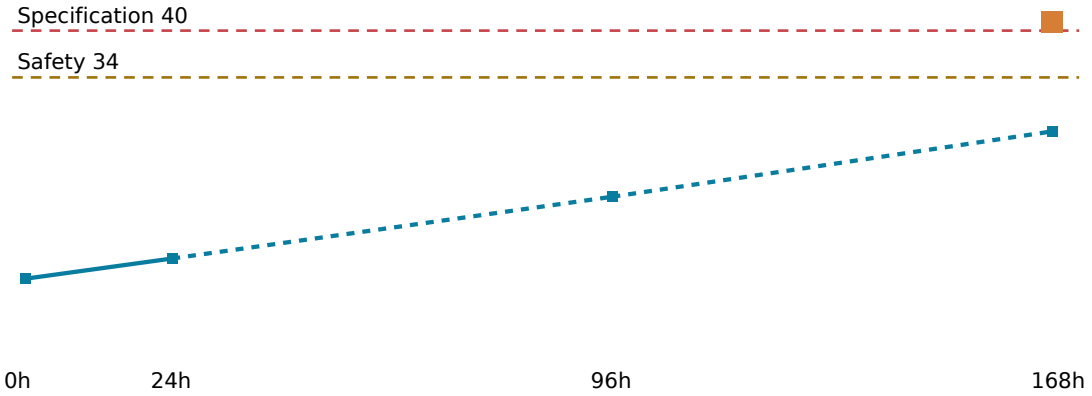
Population anomaly: robust Z=10.86; peer median=2.11 mA, MAD=0.054. Abnormal drift: peer-relative drift Z=18.29; observed change=0.838 mA. Isolation Forest score is at training-reference percentile 97.4; this is not a failure probability.




Solid: observed to screening stage. Dashed: retrospective future actual. Orange square: 168h prediction.

## Propagation delay (ns) - CRITICAL

Population anomaly: robust Z=6.41; peer median=8.43 ns, MAD=0.247. Abnormal drift: peer-relative drift Z=14.27; observed change=2.59 ns. Predicted safety crossing: 168h estimate=41 ns; upper safety threshold=34; margin=-7.04.




Solid: observed to screening stage. Dashed: retrospective future actual. Orange square: 168h prediction.

## Measurements

| parameter | unit | value 0h | value 24h | predicted 168h | risk score | status |
|---|---|---|---|---|---|---|
| Leakage current | uA | 10.672 | 15.796 | 44.599 | 93.16 | CRITICAL |
| Standby current | mA | 2.1448 | 2.9829 | 8.3476 | 62.78 | HIGH RISK |
| Propagation delay | ns | 8.1918 | 10.782 | 41.044 | 82.7 | CRITICAL |

## Methodology and limitations

Robust peer statistics, early-feature Isolation Forest and lot-disjoint regression support transparent review rules. Scores are not failure probabilities. Calibration intervals are approximate under lot dependence. Synthetic evidence is not production validation. Review states do not alter engineering measurements or decisions.